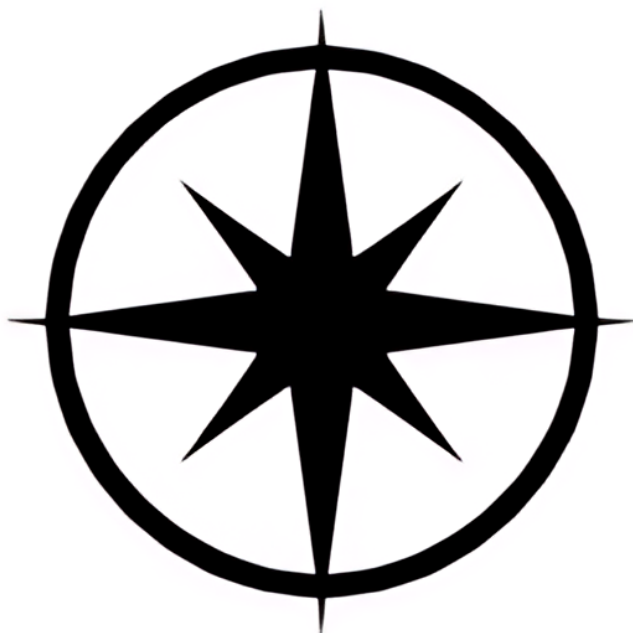




if $3n+5$ is even then n is odd

n is even then $3n+5$ is odd

$$n = 2k$$

$$3(2k+5) = 6k+15$$

$$6k + 14 + 1$$

$$2(\underbrace{3k+7}_k) + 1$$

$$2k+1 \checkmark$$

$$n(2n-1)$$

$$k(2k-1)$$

$$\text{goal: } k+1(2k+1) = 2k^2 + 3k + 1$$

$$4n-3 \quad n=k+1$$

$$= 4k+1$$

$$k(2k-1) + 4k+1$$

$$2k^2 - k + 4k + 1$$

$$2K^2 + 3K + 1$$

$$(K+1)(2K+1) \checkmark$$

$$\frac{6^n - 1}{5}$$

$$6^K - 1 = 5q$$

$$\begin{aligned} &6^{K+1} - 1 \\ &= 6 \cdot 6^K - 1 \end{aligned}$$

$$6^K = 5q + 1$$

$$6(5q + 1) - 1$$

$$30q + 6 - 1$$

$$30q + 5$$

$$5(6q + 1) = 5m \checkmark$$

no smallest positive rational number

???

if x is irrational then $\frac{x}{4}$ is irrational

$\frac{x}{4}$ is rational then x is rational

$$\frac{x}{4} = \frac{p}{q}$$

$$x = 4\left(\frac{p}{q}\right)$$

since $\frac{p}{q}$ is rational and 4 is

an integer then x is rational
original statement is true

$n^2 + n + 3$ is odd

$n^2 + n + 3$ 2 cases n is even

and n is odd

$$n = 2K$$

$$(2K)^2 + 2K + 3$$

$$4K^2 + 2K + 2 + 1$$

$$2(2K^2 + K + 1) + 1$$

$$2m + 1 \quad \text{odd}$$

$$n = 2K + 1$$

$$(2K + 1)^2 + 2K + 1 + 3$$

$$4K^2 + 4K + 1 + 2K + 1 + 3$$

$$4K^2 + 6K + 5$$

$$4K^2 + 6K + 4 + 1$$

$$2(2K^2 + 3K + 2) + 1$$

$$2n + 1 \quad \text{also odd}$$

$n^2 + n + 41$ is prime find one counter example

a counter example can be

$$n(n+1) + 41$$

$$n+1 = 40$$

$$\underline{\underline{n = 40}}$$

if $\frac{b}{a}$ and $\frac{c}{a}$ then $\frac{b+c}{a}$

$$b = aq \text{ and } c = aq$$

$$\text{then } b+c = aq$$

$$aq + aq = a(2q)$$

$$a(m) \checkmark$$

$$2^n \geq n+1$$

$$2^k \geq k+1$$

$$2^{k+1} \geq k+1+1$$

$$2 \cdot 2^k \geq k+2$$

$$2^k \geq k \quad k \geq 1$$

$$2^1 \geq 1 \quad 2 \geq 1 \quad \text{hence true}$$

$$3x+5y=1$$

$$x=-8 \quad y=5$$

